STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 13

RANDOM VARIABLES

JOINT DISCRETE RANDOM VARIABLES

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RANDOM VARIABLES

A random variable is formed by assigning a **numerical value** to each outcome in the sample space of a particular experiment. The state space of the random variable consists of these numerical values. Technically, a random variable can be thought of as being generated from a function that maps each outcome in a particular sample space onto the real number line R.

A random variable is a variable whose value is determined by the outcome of a random experiment.

*A **random variable** is a function that associates a real number with each element in the sample space.*

A **random variable** is a real-valued function whose domain is a sample space. ▪

Random variables are usually denoted by Capital letters *X, Y, Z* etc and their values are denoted by small letters *x, y, z* etc.

For example, the sample space giving a detailed description of each possible outcome when three electronic components are tested may be written

S = {NNN, NND, NDN, DNN, NDD, DND, DDN, DDD}

where *N* denotes non-defective and *D* denotes defective. One is naturally concerned with the number of defectives that occur. Thus, each point in the sample space will be *assigned a numerical value* of 0, 1, 2, or 3. These values are, of course, random quantities *determined by the outcome of the experiment*. They may be viewed as values assumed by the *random variable X*, the number of defective items when three electronic components are tested.

There are two types of random variables

1. Discrete Random Variable
2. Continuous Random Variable

DISCRETE RANDOM VARIABLE A random variable that assumes countable values is called discrete random variable.
For Example
1. Number of Students in a class
2. Number of Fish caught on a fishing trip

CONTINUIOUS RANDOM VARIABLE A random variable that can assume any value contained in one or more intervals is called continuous random variable. For Example
1. Weight of a Person
2. Temperature at a Place

JOINT PROBABILITY FNS

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JOINT PROBABILITY DISTRIBUTION

Let X and Y be two discrete random variables where X can assume the values $X: x_1, x_2, \ldots, x_m$ while Y can assume values $Y: y_1, y_2, \ldots, y_n$. Then the Probability of the event $X = x_i$ and $Y = y_j$, given by

$$P(X=x_i, Y=y_j) = f(x_i, y_j) \text{ or } P(X=x, Y=y) = f(x,y)$$

is called joint probability function of X and Y. If Note that Joint prob. fun. has the following properties

1- $f(x,y) \geq 0$ for all x, y

2- $\sum_{j=1}^{n} \sum_{i=1}^{m} f(x_i, y_j) = 1$ or $\sum_{x} \sum_{y} f(x,y) = 1$

②MARGINAL PROBABILITY FUNCTIONS

If X and Y are discrete Random variables and $f(x,y)$ is their joint prob. function the Marginal prob-fens of X and Y are defined as

Marginal Prob function for X

$$g(x_i) = \sum_{j=1}^{n} f(x_i, y_j)$$

Marginal Prob function for Y

$$h(y_j) = \sum_{i=1}^{m} f(x_i, y_j)$$

①JOINT PROBABILITY DISTRIBUTION (TABLE)

The table of joint prob fun. $f(x,y)$ is given by

X \ Y	y_1	y_2	.....	y_n	$g(x_i)$
x_1	$f(x_1,y_1)$	$f(x_2,y_2)$	---	$f(x_1,y_n)$	$g(x_1)$
x_2	$f(x_2,y_1)$	$f(x_2,y_2)$	---	$f(x_2,y_n)$	$g(x_2)$
⋮	⋮	⋮		⋮	⋮
x_m	$f(x_m,y_1)$	$f(x_m,y_2)$	----	$f(x_m,y_n)$	$g(x_m)$
$h(y_j)$	$h(y_1)$	$h(y_2)$	....	$h(y_n)$	1

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③ CONDITIONAL PROBABILITY

If x and Y are discrete random variables then the conditional prob of x given Y is defined as

$$P(X=x/Y=y) = f(x/y) = \frac{f(x,y)}{h(y)}$$

Similarly

$$P(Y=y/X=x) = f(y/x) = \frac{f(x,y)}{g(x)}$$

④ INDEPENDENCE OF TWO RANDOM VARIABLES

Then Random variables x and Y of the joint probability function $f(x,y)$ are statistically independent random variables iff

$$f(x,y) = g(x).h(y)$$

⑤ JOINT DISTRIBUTION FUNCTION.

If x and Y are discrete random variables then the joint distribution function of x and Y is defined by

$$F(x,y) = P(X \le x, Y \le y) = \sum_{u \le x} \sum_{v \le y} f(u,v)$$

EXAMPLES

EXAMPLE① Two marbles are selected at random from a Box containing 3 black, 2 Red and 3 green marbles. If X is the number of black marbles and Y is the number of red marbles selected then find the following

(a) Joint Probability Function.
(b) Marginal Probability functions
(c) $P(X+Y \leq 1)$
(d) conditional Probability functions $f(x/1)$ and $f(2/y)$
(e) $P(X=0/Y=1)$
(f) Are X and Y independent.

SOLUTION Here Black = 3, Red = 2, Green = 3

(a) Total = 8, Select = 2, $n(S) = {}^8C_2 = 28$

Also $X = 0,1,2$ and $Y = 0,1,2$ as X: Black, and Y: Red.

Construct Table

X \ Y	0	1	2	g(x)
0	3/28	6/28	1/28	10/28
1	9/28	6/28	0	15/28
2	3/28	0	0	3/28
h(y)	15/28	12/28	1/28	1

Note in above table

$$f(0,0) = \frac{{}^3C_0 \times {}^2C_0 \times {}^3C_2}{{}^8C_2}, \quad f(0,1) = \frac{{}^3C_0 \times {}^2C_1 \times {}^3C_1}{{}^8C_2}, \quad f(0,2) = \frac{{}^3C_0 \times {}^2C_2 \times {}^3C_0}{{}^8C_2}$$

$$f(1,0) = \frac{{}^3C_1 \times {}^2C_0 \times {}^3C_1}{{}^8C_2}, \quad f(1,1) = \frac{{}^3C_1 \times {}^2C_1 \times {}^3C_0}{{}^8C_2}, \quad f(1,2) = \frac{0}{{}^8C_2}$$

$$f(2,0) = \frac{{}^3C_2 \times {}^2C_0 \times {}^3C_0}{{}^8C_2}, \quad f(2,1) = \frac{0}{{}^8C_2}, \quad f(2,2) = \frac{0}{{}^8C_2}$$

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(b) Marginal Prob. function of x and y are given as

x	0	1	2
g(x)	10/28	15/28	3/28

y	0	1	2
h(y)	15/28	12/28	1/28

(c) $P(X+Y \le 1)$

$= f(0,0) + f(0,1) + f(1,0)$

$= \frac{3}{28} + \frac{6}{28} + \frac{9}{28}$

$= \frac{18}{28} = \frac{9}{14}$

(d) conditional Prob functions

For $f(x/1)$

we have to find

$f(0/1) = \frac{f(0,1)}{h(1)} = \frac{6/28}{12/28} = \frac{1}{2}$

$f(1/1) = \frac{f(1,1)}{h(1)} = \frac{6/28}{12/28} = \frac{1}{2}$

$f(2/1) = \frac{f(2,1)}{h(1)} = \frac{0}{12/28} = 0$

Also $f(2/y)$

we have to find

$f(2/0) = \frac{f(2,0)}{h(0)} = \frac{3/28}{15/28} = \frac{3}{15} = \frac{1}{5}$

$f(2/1) = \frac{f(2,1)}{h(1)} = \frac{0}{12/28} = 0$

$f(2/2) = \frac{f(2,2)}{h(2)} = \frac{0}{1/28} = 0$

(e) $P(X=0/Y=1) = f(0/1) = \frac{1}{2}$

(f) For independence we check $f(x,y) = g(x).h(y)$ (Nine times)

Now

$f(0,0) = g(0)h(0)$

$\frac{3}{28} \neq \frac{10}{28} \times \frac{15}{28}$

$\frac{3}{28} \neq \frac{75}{392}$ so x and y are not independent.

EXAMPLE-② Flip a coin three times and record the number of heads appeared. Let 'x' denote the number of heads on the last (third) Flip. and 'y' denote the total number of heads in three flips. Then find the following

(a) Joint Probability function.

(b) Marginal Probability Functions.

(c) $P(X<Y)$

(d) conditional Probabilities $f(x|2)$ and $f(1/y)$

(e) Are X and Y are independent?

SOLUTION (a) Here $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

Also X takes the values $\{0,1\}$

and Y takes the values $\{0,1,2,3\}$

		Y				
		0	1	2	3	$g(x)$
X	0	1/8	2/8	1/8	0	4/8
	1	0	1/8	2/8	1/8	4/8
	$h(y)$	1/8	3/8	3/8	1/8	1

which is the required probability function of X & Y.

(b) Marginal Probability Functions

X	0	1
$g(x)$	4/8	4/8

Y	0	1	2	3
$h(y)$	1/8	3/8	3/8	1/8

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(c) $P(X<Y)$

$= f(0,1) + f(0,2) + f(0,3) + f(1,2) + f(1,3)$

$= \frac{2}{8} + \frac{1}{8} + 0 + \frac{2}{8} + \frac{1}{8}$

$= \frac{6}{8} = \frac{3}{4}$

(d) $f(x/2)$

$f(0/2) = \frac{f(0,2)}{h(2)} = \frac{1/8}{3/8} = \frac{1}{3}$

$f(1/2) = \frac{f(1,2)}{h(2)} = \frac{2/8}{3/8} = \frac{2}{3}$

$f(1/y)$

$f(1/0) = \frac{f(1,0)}{h(0)} = \frac{0}{1/8} = 0$

$f(1/1) = \frac{f(1,1)}{h(1)} = \frac{1/8}{3/8} = 1/3$

$f(1/2) = \frac{f(1,2)}{h(2)} = \frac{2/8}{3/8} = \frac{2}{3}$

$f(1/3) = \frac{f(1,3)}{h(3)} = \frac{1/8}{1/8} = 1$

(e) Independence.

For independence $f(x,y) = g(x).h(y)$ we check all elements one by one.

$f(0,0) \neq g(0).h(0)$

$\frac{1}{8} \neq (\frac{4}{8})(\frac{1}{8})$

$\Rightarrow$ x and y are not independent.

EXAMPLE (3) of the joint Prob. distribution of x and y is given by $f(x,y) = \frac{x+y}{30}$ for $x = 0,1,2,3$; $y = 0,1,2$

Then find

(a) Marginal Prob functions of x and y

(b) $P(X \leq 2, Y = 1)$

(c) $P(X > 2, Y \leq 1)$

(d) $P(X > Y)$

(e) $P(X + Y = 4)$

SOLUTION Joint Prob distribution is given in the table as

x \ y	0	1	2	g(x)
0	0	1/30	2/30	3/30
1	1/30	2/30	3/30	6/30
2	2/30	3/30	4/30	9/30
3	3/30	4/30	5/30	12/30
h(y)	6/30	10/30	14/30	1

(a) Marginal Prob for X

x	0	1	2	3
g(x)	$\frac{3}{30}$	$\frac{6}{30}$	$\frac{9}{30}$	$\frac{12}{30}$

Marginal Prob for for Y

y	0	1	2
h(y)	$\frac{6}{30}$	$\frac{10}{30}$	$\frac{14}{30}$

(b) $P(X \leq 2, Y = 1) = f(0,1) + f(1,1) + f(2,1) = \frac{1}{30} + \frac{2}{30} + \frac{3}{30} = \frac{1}{5}$

(c) $P(X > 2, Y \leq 1)$

$= f(3,1) + f(3,0) = \frac{4}{30} + \frac{3}{30} = \frac{7}{30}$

(d) $P(X > Y)$

$= f(1,0) + f(2,0) + f(2,1) + f(3,0) + f(3,1) + f(3,2)$

$= \frac{1}{30} + \frac{2}{30} + \frac{3}{30} + \frac{3}{30} + \frac{4}{30} + \frac{5}{30} = \frac{18}{30} = \frac{3}{5}$

(e) $P(X + Y = 4) = f(2,2) + f(3,1) = \frac{4}{30} + \frac{4}{30} = \frac{8}{30} = \frac{4}{15}$

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EXPECTATION JOINT DRV'S

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EXPECTATION OF JOINT DISCRETE RANDOM VARIABLES

DEFINITION Let X and Y be two Discrete Random Variables with a joint probability distribution $f(x,y)$. Then Mean or expected value of random variable $g(x,y)$ is given by

$$E[g(x,y)] = \sum_x \sum_y g(x,y) f(x,y)$$

Note that $E[g(x,y)] = \mu_{g(x,y)}$

SOME IMPORTANT FORMULAS AND CONCEPTS

1- If X and Y are Random variables, then

$$E(X \pm Y) = E(X) \pm E(Y)$$

2- If X and Y are independent Random Variables, then

$$E(XY) = E(X).E(Y)$$

3- If joint distribution of X and Y is given then.

$$E(X) = \sum x g(x).$$

and $E(Y) = \sum y h(y)$

4- (a) $E(X+Y) = \sum\sum (x+y) f(x,y)$

(b) $E(X-Y) = \sum\sum (x-y) f(x,y)$

(c) $E(XY) = \sum\sum xy\, f(x,y)$

(d) $E\left(\frac{X}{Y}\right) = \sum\sum \left(\frac{x}{y}\right) f(x,y)$

5- ### COVARIANCE OF X and Y

If X and Y are random variables with joint Prob-distribution $f(x,y)$. Then covariance of X and Y is given by

$$\sigma_{xy} = Cov(x,y) = E(XY) - E(X).E(Y)$$

6- ### CORRELATION COEFFICIENT

If X and Y are RVs with covariance σ_{xy} and standard deviations σ_x and σ_y respectively. Then correlation coefficient of X and Y is given by $\boxed{\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}}$ $-1 \le \rho \le 1$

EXAMPLES

EXAMPLE-① Let X and Y be two discrete random variables with the following probability distribution function

Y \ X	1	2	3
1	0.03	0.04	0.03
2	0.15	0.20	0.15
3	0.12	0.16	0.12

Find
(a) E(X) (b) E(Y) (c) E(X+Y) (d) E(XY) (e) check whether X and Y are independent or not.

SOLUTION

Y \ X	1	2	3	h(y)
1	0.03	0.04	0.03	0.1
2	0.15	0.20	0.15	0.5
3	0.12	0.16	0.12	0.4
g(x)	0.3	0.4	0.3	1

Now

(a) $E(x) = \sum x g(x) = 1(0.3) + 2(0.4) + 3(0.3) = 2$

(b) $E(y) = \sum y h(y) = 1(0.1) + 2(0.5) + 3(0.4) = 2.3$

(c) $E(X+Y) = \sum\sum (x+y) f(x,y)$

$= (1+1)(0.03) + (1+2)(0.15) + (1+3)(0.12) + (2+1)(0.04) + (2+2)(0.20) + (2+3)(0.16) + (3+1)(0.03) + (3+2)(0.15) + (3+3)(0.12)$

$= 4.3$

(d) $E(XY) = \sum\sum xy f(x,y) = (1)(1)(0.03) + (1)(2)(0.15) + (1)(3)(0.12) + (2)(1)(0.04) + (2)(2)(0.20) + (2)(3)(0.16) + (3)(1)(0.03) + (3)(2)(0.15) + (3)(3)(0.12)$

$= 4.6$

(e) Now $E(xy) = E(x).E(y)$

$4.6 = (2)(2.3)$

$4.6 = 4.6$

So x and y are independent R.V's.

EXAMPLE-② If x and y are DRV's having joint Prob. distribution given below.

x \ y	0	1	2
0	3/28	6/28	1/28
1	9/28	6/28	0
2	3/28	0	0

Find

(a) $E(x)$ (b) $E(y)$ (c) $E(x+y)$ (d) $E(xy)$ (e) check independence.

SOLUTION

x \ y	0	1	2	g(x)
0	3/28	6/28	1/28	10/28
1	9/28	6/28	0	15/28
2	3/28	0	0	3/28
h(y)	15/28	12/28	1/28	1

(a) $E(x) = \sum x g(x) = 0(\frac{10}{28}) + 1(\frac{15}{28}) + 2(\frac{3}{28}) = \frac{3}{4}$

(b) $E(y) = \sum y h(y) = 0(\frac{15}{28}) + 1(\frac{12}{28}) + 2(\frac{1}{28}) = \frac{1}{2}$

(c) $E(x+y) = \sum\sum (x+y) f(x,y)$

$= (0+0)(\frac{3}{28}) + (0+1)(\frac{6}{28}) + (0+2)(\frac{1}{28}) + (1+0)(\frac{9}{28}) + (1+1)(\frac{6}{28})$
$+ (1+2)(0) + (2+0)(3/28) + (2+1)(0) + (2+2)(0) = 5/4$

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(d) $E(XY) = \sum\sum xy f(x,y)$

$= (0)(0)(\frac{3}{28}) + (0)(1)(\frac{6}{28}) + (0)(2)(\frac{1}{28}) + (1)(0)(\frac{9}{28})$

$+ (1)(1)(\frac{6}{28}) + (1)(2)(0) + (2)(0)(\frac{3}{28}) + (2)(1)(0)$

$+ (2)(2)(0)$

$= \frac{3}{14}$

$\Rightarrow \boxed{E(XY) = \frac{3}{14}}$

(e) INDEPENDENCE

Now $E(XY) = \frac{3}{14}$

$E(X) = \frac{3}{4}$

$E(Y) = \frac{1}{2}$

So $E(X) \cdot E(Y) = (\frac{3}{4})(\frac{1}{2}) = \frac{3}{8}$

So $E(XY) \neq E(X) \cdot E(Y)$

$\Rightarrow$ X and Y are not independent Random variables

COVARIANCE AND CORRELATION

COVARIANCE If x and y are random variables with joint probability distribution/density function $f(x,y)$ then covariance of x and y is given by

$$COV(x,y) = \sigma_{xy} = E[(x-E(x))(y-E(y))]$$

or $$COV(x,y) = \sigma_{xy} = E(xy) - E(x)\,E(y)$$

PROPERTIES OF COVARIANCE.

① If x and y are discrete Random Variables then

$$COV(x,y) = \sum_x \sum_y (x-E(x))(y-E(y))$$

② If x and y are continuous Random variables then

$$COV(x,y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x-E(x))(y-E(y))\,dy\,dx$$

③ If x and y are independent Random variables then $COV(x,y) = 0$

④ $COV(x,x) = E[(x-E(x))(x-E(x))]$

$$= E(x-E(x))^2 = Var(x)$$

⑤ If x and y are two random variables then

(a) $Var(x+y) = Var(x) + Var(y) + 2\,COV(x,y)$

(b) $Var(x-y) = Var(x) + Var(y) - 2\,COV(x,y)$

CORRELATION

If x and y are random variables with covariance 'σ_{xy}' and standard deviations σ_x and σ_y respectively. Then correlation coefficient of x and y is given by

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x . \sigma_y}$$

Correlation measures the degree of linear Association between the two random variables. It is not influenced by the units used to measure the variables.

PROPERTIES OF CORRELATION CO-EFFICIENT

① The correlation between two variables is bounded by -1 and 1. ie $-1 \leq \rho_{xy} \leq 1$

② $|\rho_{xy}| = 1$ iff $y = a + bx$. Thus, the correlation between two variables will equal to 1 in absolute value only if there is a perfect linear relationship between the two variables.

③ correlation does not imply causality. Two variables may move together but not be causally related (For eg: shoe size and strength are probably positively related but buying a larger shirt will probably not make you stronger.

EXAMPLES

EXAMPLE ⑥ If 'x' and 'y' are Discrete Random Variables as defined in Example ①. Then Find.

(a) σ_{xy} (covariance)

(b) ρ_{xy} (correlation coefficient).

SOLUTION

(a) COVARIANCE.

$$\sigma_{xy} = E(xy) - E(x).E(y)$$

$$= (4.6) - (2)(2.3)$$

$$\sigma_{xy} = 4.6 - 4.6 = 0 \qquad \text{So } \boxed{\sigma_{xy} = 0}$$

(b) CORRELATION

For ρ_{xy} we need to find σ_{xy}, σ_x, σ_y

Now
$$\sigma_x = \sqrt{Var(x)} = E(x^2) - (E(x))^2$$

Now
$$E(x^2) = (1)^2(0.3) + (2)^2(0.4) + (3)^2(0.3) = 4.6$$

Also $E(x) = 2$

So $Var(x) = E(x^2) - (E(x))^2 = 4.6 - (2)^2 = 0.6$

$\Rightarrow \quad \sigma_x = \sqrt{0.6}$

Also $\sigma_y = \sqrt{Var(y)}$ and $Var(y) = E(y^2) - (E(y))^2$

Now $E(y^2) = (1)^2(0.1) + (2)^2(0.5) + (3)^2(0.4) = 5.7$

Also $E(y) = 2.3$

So $Var(y) = E(y^2) - (E(y))^2 = 5.7 - (2.3)^2 = 0.41$

So $\sigma_y = \sqrt{0.41}$

Now $\rho_{xy} = \dfrac{\sigma_{xy}}{\sigma_x . \sigma_y} = \dfrac{0}{\sqrt{0.6 \times 0.41}} = 0$ So $\boxed{\rho_{xy} = 0}$

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EXAMPLE-⑦ If x and y are DRV as defd in Exp②. Then Find (a) σ_{xy} (b) ρ_{xy}

(a) Covariance of x and y ie σ_{xy}

Now $\sigma_{xy} = E(xy) - E(x).E(y)$

$= \left(\frac{3}{14}\right) - \left(\frac{3}{4}\right)\left(\frac{1}{2}\right)$

$\sigma_{xy} = -\frac{9}{56}$

(b) correlation coefficient of x and y ie ρ_{xy}

WKT $\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x . \sigma_y}$

Now we first find σ_x, σ_y and σ_{xy}

For $\sigma_x = \sqrt{Var(x)}$ also $Var(x) = E(x^2) - (E(x))^2$

Now $E(x^2) = (0)^2\left(\frac{10}{28}\right) + (1)^2\left(\frac{15}{28}\right) + (2)^2\left(\frac{3}{28}\right) = \frac{27}{28}$, Also $E(x) = \frac{3}{4}$

and $Var(x) = E(x^2) - (E(x))^2 = \frac{27}{28} - \left(\frac{3}{4}\right)^2 = \frac{45}{112}$

do $\sigma_x = \sqrt{Var(x)} = \sqrt{\frac{45}{112}}$

Now $\sigma_y = \sqrt{Var(y)}$ also $Var(y) = E(y^2) - (E(y))^2$

$E(y^2) = (0)^2\left(\frac{15}{28}\right) + (1)^2\left(\frac{12}{28}\right) + (2)^2\left(\frac{1}{28}\right) = \frac{4}{7}$, Also $E(y) = \frac{1}{2}$

do $Var(y) = E(y^2) - (E(y))^2 = \frac{4}{7} - \left(\frac{1}{2}\right)^2 = \frac{9}{28}$

& $\sigma_y = \sqrt{\frac{9}{28}}$

Now $\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x . \sigma_y} = \frac{(-9/56)}{\sqrt{\frac{45}{112}} . \sqrt{\frac{9}{28}}} = -\frac{1}{\sqrt{5}} = -0.44$

do $\boxed{\rho_{xy} = -0.44}$

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